Resilient AI in Therapeutic Rehabilitation: The Integration of Computer Vision and Deep Learning for Dynamic Therapy Adaptation

Team Members

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Agenda

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- Literature Survey
- Outcome of Literature Survey
- Problem Statement and Objectives
- Existing Methodology
- Dataset details
- Proposed enhancements/novelty
- Work done (Current status)
- Individual Contribution

Introduction

The Healthcare Challenge

- 2.4 billion people globally require rehabilitation services
- Only 50% of affected populations have access to rehabilitation
- Physiotherapist shortage in rural and underserved areas
- Manual monitoring is time-intensive and inconsistent

The Solution

- AI-powered automated exercise monitoring
- Real-time feedback without constant clinician supervision
- Objective, quantifiable assessment of movement quality
- Scalable system deployable on low-cost edge devices

Introduction

Target Application

- Home-based rehabilitation programs
- Post-surgical recovery monitoring
- Sports injury rehabilitation
- Elderly care and mobility assistance

Literature Review

Author/year	Methodology	Advantages	Limitations
Cirillo, E., et al. 2025	• Combines Kinect 3D tracking and MoveNet for accurate real-time joint detection. • Uses reinforcement learning to dynamically adjust therapy difficulty and assistance levels.	• Enables real-time, markerless rehabilitation monitoring, enhancing patient accessibility and comfort. • Delivers accurate posture correction and personalized feedback via deep learning analysis.	• Performance accuracy degrades due to occlusions, lighting variations, or environmental disturbances. • Requires therapist supervision as AI adjustments are recommendations, not autonomous decisions.
Atima T. et al. 2024	• Surveyed 45 articles using PRISMA to analyze pose estimation, assessment, and feedback methods. • Categorized systems by pose libraries, assessment techniques, and feedback modalities via bottom-up analysis.	• Highlights libraries like OpenPose and MediaPipe, enabling real-time, accessible consumer applications. • Identifies diverse mathematical, rule-based, and ML assessment methods informing robust feedback designs.	• Studies lack detailed descriptions of feedback content, timing, and interface, hindering reproducibility. • Evaluation methods vary widely with limited standardized datasets, reducing assessment performance comparability.

Literature Review

Author/year	Methodology	Advantages	Limitations
Liao et al. 2020	• Uses autoencoder-based dimensionality reduction and GMM log-likelihood for performance metric. • Spatio-temporal deep network with temporal pyramids, multi-branch convolutions and LSTM layers.	• GMM metric better separates correct and incorrect movements after reduction. • Model outperforms several baseline architectures; fast inference around ten milliseconds.	• Validated mainly on healthy-subject datasets captured by expensive Vicon system. • Limited clinical labels and patient data; requires clinician-annotated scores for robustness.
Baba et al. 2012	• Used Kinect depth camera and skeleton tracking to extract 3D orientation features. • Trained temporal multi-class classifiers (HM-SVM, HM-SVM◦) with three-stage active labeling.	• Real-time compensation detection achieved using inexpensive consumer depth sensor hardware. • Active learning drastically reduced manual annotation while preserving classifier performance.	• Dataset used healthy subjects simulating compensations, significantly limiting clinical generalization. • Ambiguous labels near transition boundaries reduce temporal segmentation and evaluation reliability.

Outcome of Literature Review

- Computer vision enables objective, continuous kinematic monitoring, replacing subjective clinical observation during home and clinic rehabilitation.
- Deep temporal models (CNN-LSTM, 3D-CNN) better capture sequential motion, detecting compensations and fatigue accurately.
- Adaptive feedback loops integrating perception, decision, and control enable dynamic therapy adjustments in real time.
- Data scarcity, variable capture setups, and simulated datasets limit generalization across patients and clinical settings

Problem Statement

- Traditional physiotherapy rehabilitation requires continuous supervision by trained clinicians to ensure patients perform exercises correctly. This creates accessibility barriers, scalability issues, consistency problems and cost burden

Research Objectives

1. Real-time pose estimation
2. Automated correctness classification
3. Confidence-based reliability
4. Performance metrics computation
5. Scalable architecture

Existing Methodology

- **Mathematical Formulation**

$$C = ||Y_{correct} - Y_{actual}|| = \sqrt{\sum_{i=1}^{17} (y_{correct,i} - y_{actual,i})^2}$$

- **Root Mean Squared Error (RMSE)**

$$RMSE = \sqrt{MSE} = \sqrt{\frac{1}{N} \sum_{i=1}^{N} (y_{correct,i} - y_{actual,i})^2}$$

Existing Methodology

- **Confidence Scoring**

  Threshold-Based Decision

  Confidence ≥ 0.8: Prediction is **reliable** → Proceed to classification

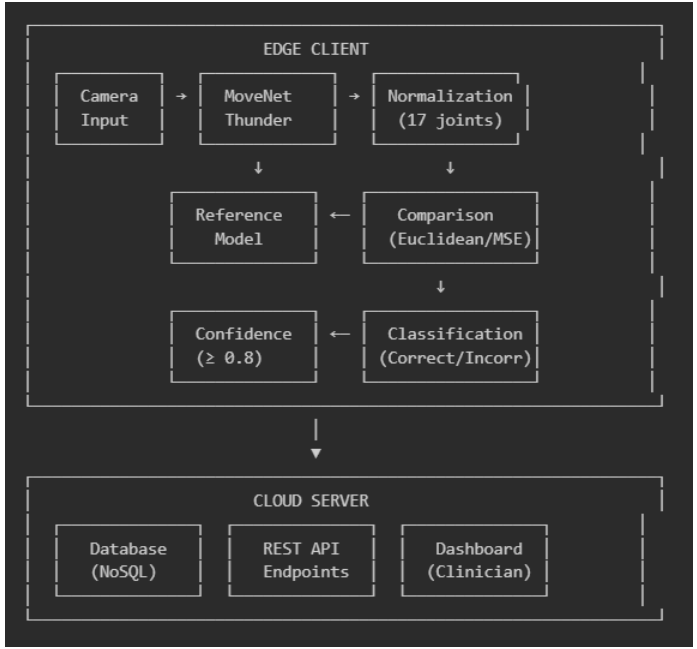
  Confidence < 0.8: Prediction is **uncertain** → Display warning to user

Classification

RMSE vs Threshold	Confidence	Result
Low RMSE	≥ 0.8	**Correct Execution** (Green)
High RMSE	≥ 0.8	**Incorrect Execution** (Blue)
Any	< 0.8	**Warning** (Unreliable detection)

Proposed enhancements/novelty

- High Level Overview:



Proposed enhancements/novelty

- The base research paper provides core methodology. Our implementation extends this with:

Enhancement Area	Research Paper	Our Extension
Architecture	Monolithic	**Edge-Cloud Distributed**
Communication	Direct	**Publisher-Subscriber Model**
Missing Data	No handling	**Joint Imputation Module**
Error Detection	Basic	**Compensation Detection**
Temporal Analysis	Frame-by-frame	**LSTM Fluidity Analysis**
Data Management	Not specified	**NoSQL with XML Format**

Proposed enhancements/novelty

Team Distribution:

Sub team 1: Siddharth and Srujan

- Joint Imputation
- Compensation Detection
- LSTM Fluidity Analysis

Sub team 2: Anirudh and Siva Sai

- Publisher-Subscriber Model
- NoSQL Database + XML Format
- REST API Development

Dataset Details / Statistics

- **REHAB24-6 Dataset**

Attribute	Value
Dataset Name	REHAB24-6
Total Exercises	6 different rehabilitation exercises
Participants	Multiple subjects (varying body types)
Camera Views	2 cameras (c17, c18)
Frame Rates	30 fps, 120 fps
Data Types	2D Joints, 3D Joints, 2D Markers, 3D Markers, Videos

Work done (Current status of work)

Module 1: MoveNet Pose Estimation

Implementation:

- TensorFlow Hub integration for MoveNet Thunder model
- 17 keypoint extraction with (x, y, confidence)
- Video frame processing pipeline
- Fallback mode for CPU-only systems

Module 2: Skeleton Normalization

Implementation:

- Hip center calculation (midpoint of left/right hip)
- Translation to origin

Work done (Current status of work)

$$P_{normalized} = \frac{P_{original} - Hip_{center}}{Torso_{length}}$$

Module 3: Reference Model Building

Implementation:

- Load correct execution samples from dataset
- Compute mean joint positions across correct samples
- Calculate standard deviation for variability bounds
- Store reference template for comparison

Work done (Current status of work)

Module 4: Movement Comparison

Implementation:

- Euclidean distance computation between observed and reference
- Per-joint distance analysis for localized feedback
- MSE and RMSE calculation
- Movement deviation scoring

Module 5: Confidence Scoring & Classification

Implementation:

- Confidence scoring from RMSE values
- 0.8 threshold for reliable predictions
- Binary classification (correct = 1, incorrect = 0)
- Warning system for low-confidence predictions

Results and Analysis (As per current status)

```
============================================================
Total samples loaded: 1072
  Correct: 568
  Incorrect: 504
============================================================

Training set: 857 samples
Validation set: 215 samples

------------------------------------------------------------
```

Results and Analysis (As per current status)

```
============================================================
EVALUATION
============================================================
  Validation Loss: 0.0918
  Validation Accuracy: 97.67%

Confusion Matrix:
                   Predicted
                   Correct  Incorrect
  Actual Correct     112         0
  Actual Incorrect     5        98

Metrics (for 'Correct' class):
  Precision: 95.73%
  Recall: 100.00%
  F1 Score: 97.82%
```

Individual Contribution of each student

Sl.no	Name – Reg. no	Details
1	Anirudh S 231IT006	**Pose Estimation & Normalization**
2	Avulakonda Siva Sai 231IT012	**Reference Model & Comparison**
3	Siddharth S K 231IT071	**Confidence Scoring & Classification**
4	T Srujan Swamy 231IT079	**Data Management & API Foundation**

References

1. E. Cirillo, C. Conte, A. Moccardi and M. Fonisto,“Resilient AI in Therapeutic Rehabilitation: The Integration of Computer Vision and Deep Learning for Dynamic Therapy Adaptation,” *Applied Sciences*, vol. 15, no. 6800, 2025.
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4. A. Tharatipyakul, T. Srikaewsiew and S. Pongnumkul,“Deep Learning-Based Human Body Pose Estimation in Providing Feedback for Physical Movement: A Review,” *Heliyon*, vol. 10, 2024.